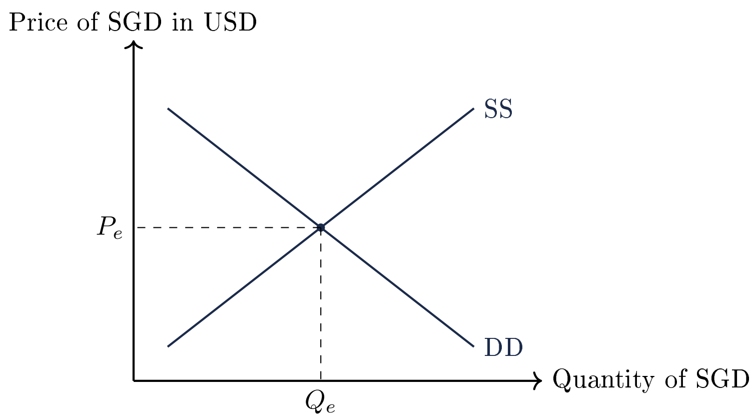
Document 17. Exchange Rate and Exchange Rate Systems

Topic overview. The exchange rate is the price of a domestic currency relative to another foreign currency. Its determination depends on demand and supply of the currency in the foreign exchange (forex) market. Demand for a currency is derived from foreigners' demand for the country's goods, services, and financial assets; supply is derived from residents' demand for foreign goods, services, and financial assets.

Section A. Important Definitions

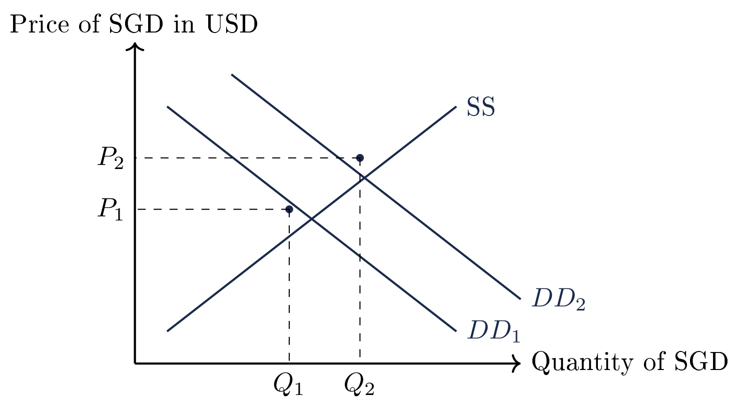
Exchange rate	Price of a domestic currency in terms of another foreign currency.
Demand for currency	Derived demand from foreigners' demand for goods, services, and financial assets.
Supply of currency	Derived from residents' demand for foreign goods, services, and financial assets.
Appreciation / Depreciation	Free-market rise / fall in value under floating/managed systems.
Revaluation / Devaluation	Government-set rise / fall under a fixed system.
Forex market	Market where the demand and supply of a currency determines its price.

Section B. Diagram — Equilibrium Exchange Rate



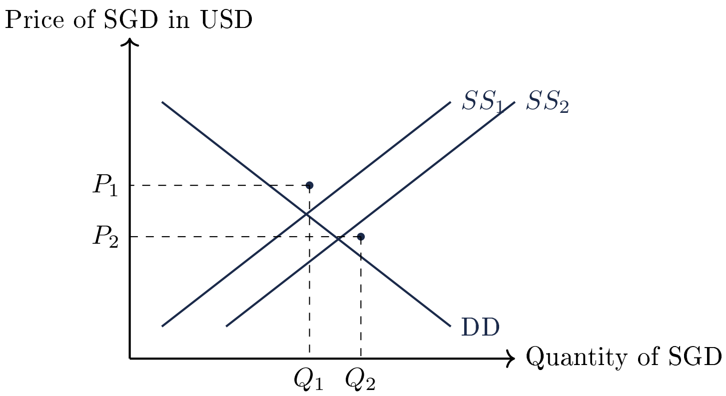
Section C. Factor 1: Demand for Exports / Imports (verbatim Crucible)

Scenario: Foreigners demand more Singapore exports



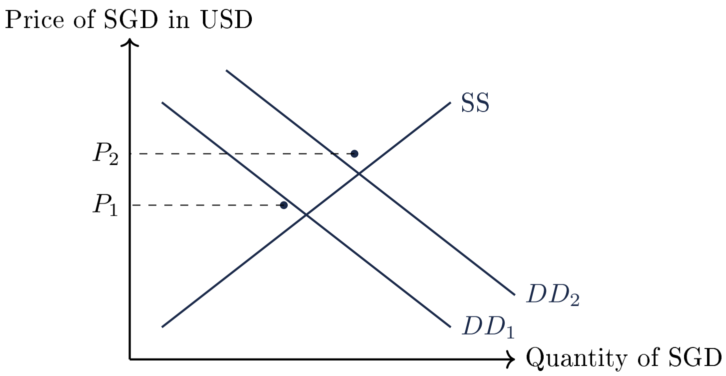
Step 1. If there is an increase in the demand for Singapore's goods and services by foreigners, this would increase the demand for Singapore's currency, as foreigners would require Singapore dollar to purchase Singapore's goods and services.
Step 2. This increases the demand of the SGD from DD_1 to DD_2, and this leads to an increase in the value of the SGD from P_1 to P_2.
Step 3. Hence, the SGD **appreciates**.

Opposite scenario: Singaporeans demand more foreign imports



Step 1. An increase in demand for foreign imports by Singaporeans would lead to an increase in demand for foreign currencies in the forex market. Singaporeans will thus have to sell their SGD in order to purchase these foreign currencies in the forex market, to purchase these foreign imports.
Step 2. This leads to an increase in supply of the SGD in the forex market from SS_1 to SS_2, and there is a decrease in the value of the SGD from P_1 to P_2.
Step 3. Hence, the SGD **depreciates**.

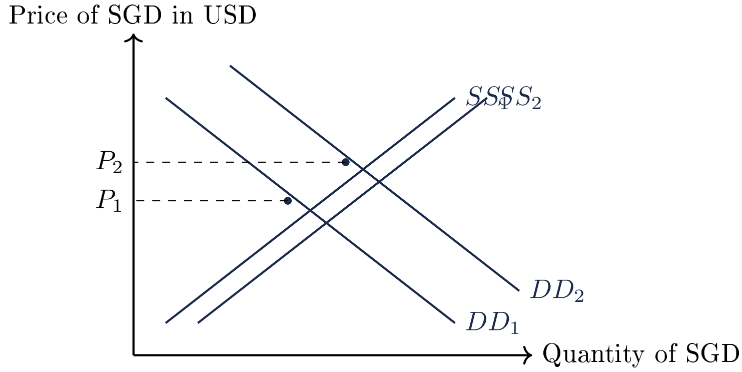
Section D. Factor 2: Movement of Capital / FDI (verbatim Crucible)



Step 1. If there is an increase in FDI into Singapore, this would lead to an increase in demand for the SGD in the forex market, as foreigners would require SGD to invest into Singapore. Similarly, if more foreigners decide to save in Singapore's banks, they would require Singapore dollar to save in Singapore banks, and this would increase the demand for Singapore dollar in the forex market.
Step 2. This would increase the demand of the SGD in the forex market from DD_1 to DD_2, which causes the value of the SGD to increase from P_1 to P_2.
Step 3. Hence, the SGD **appreciates**.

Section E. Factor 3: Changes in Relative Real Income (verbatim Crucible)

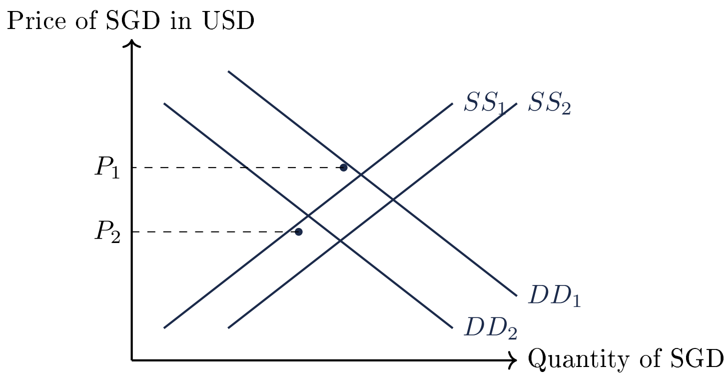
Scenario 1: UK real income rises more than Singapore's



Step 1. If UK's real income rises faster relative to Singapore's, this will increase the demand for Singapore's goods and services (assuming $YED > 0$). As demand for SGD is a derived demand of Singapore's exports, this would lead to an increase in demand for SGD.
Step 2. However, since the real income of Singapore is also rising, this would mean that Singaporeans would also demand more goods and services from the UK (assuming $YED > 0$ for UK imports). Hence, Singaporeans would now sell SGD in the forex market to purchase UK pound, leading to an increase in supply of SGD.
Step 3. However, as UK's relative real income rises faster than Singapore's, the demand for SGD would increase more from DD_1 to DD_2 than the supply of SGD from SS_1 to SS_2. The value of SGD increases from P_1 to P_2.
Step 4. Hence, the SGD **appreciates**.

Section F. Factor 4: Changes in Inflation Rates (verbatim Crucible)

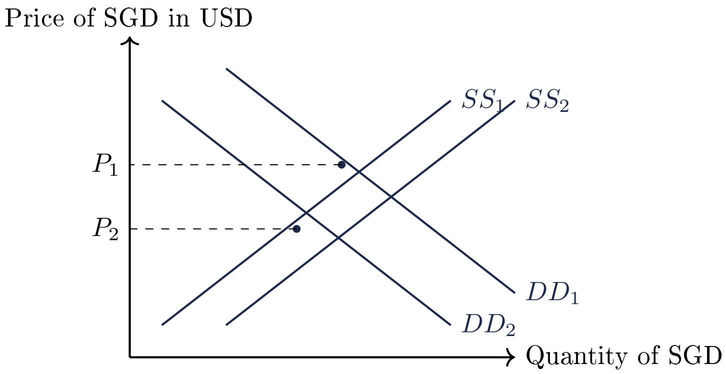
Scenario 1: SG inflation rises relative to the UK



Step 1. A rise in Singapore's inflation rate relative to the UK will cause British goods to be relatively cheaper. Singaporeans will switch to purchasing British imports (assuming British imports are close substitutes). This leads to an increase in demand for British imports; Singaporeans sell SGD in the forex market to purchase British pound. Hence, this leads to an increase in supply of SGD.
Step 2. At the same time, UK citizens will demand less of Singapore's exports as they are now relatively more expensive. Assuming $PED_x > 1$ due to wide availability of substitutes, Singapore's export revenue will decrease, and demand for SGD will decrease.
Step 3. The increase in supply from SS_1 to SS_2, and the decrease in demand from DD_1 to DD_2 causes the value of SGD to fall from P_1 to P_2.
Step 4. Hence, the SGD **depreciates**.

Section G. Factor 5: Changes in Relative Interest Rates (verbatim Crucible)

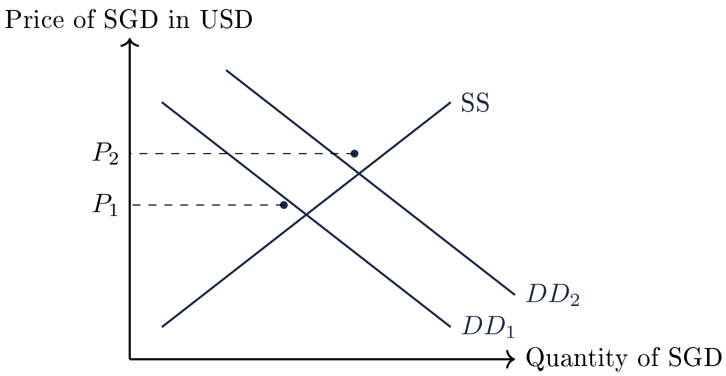
Scenario 1: Singapore's interest rates fall relative to other countries



Step 1. If the interest rates in Singapore fall relative to foreign countries, Singaporeans would be more encouraged to deposit their funds into foreign banks to earn the higher rate of interest. Singaporeans would sell SGD in the forex market to purchase foreign currencies. This leads to an increase in supply of SGD.
Step 2. Foreigners would be discouraged from depositing their funds in Singapore's banks as they now earn less interest. Foreigners would demand less SGD in the forex market.
Step 3. The increase in supply from SS_1 to SS_2 and the decrease in demand from DD_1 to DD_2 causes the value of SGD to decrease from P_1 to P_2.
Step 4. Hence, the SGD **depreciates**.

Section H. Factor 6: Speculation of Exchange Rates (verbatim Crucible)

Scenario 1: Speculators expect appreciation of SGD



Step 1. When speculators expect an appreciation of the SGD, they would want to hold onto more SGD to gain the benefit of the increase in value. Hence, speculators will sell foreign currencies in the forex market to purchase SGD.
Step 2. This leads to an increase in demand of the SGD from DD_1 to DD_2, and the value of the SGD increases from P_1 to P_2.
Step 3. Hence, the SGD **appreciates**.

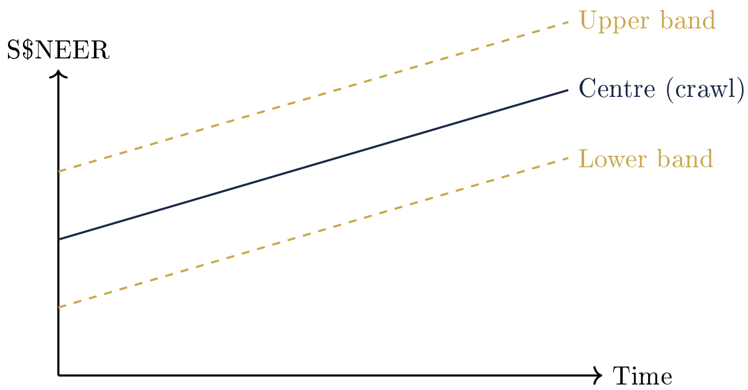
Section I. Exchange Rate Systems

(1) Freely-floating. Rate fully determined by demand and supply in the forex market. No government intervention. BOP always in equilibrium because surpluses / deficits are automatically eliminated via exchange rate adjustments. Foreign reserves never used.

(2) Fixed. Exchange rate fixed by the government; does not vary according to D & S. Government intervenes in the forex market to maintain the peg by buying / selling its own currency using reserves.

(3) Managed Float (BBC — Basket, Band, Crawl). Hybrid system. Government allows the exchange rate to fluctuate within a controlled band with an upper and lower limit. If the rate hits a band, the central bank intervenes.

Section I1. Diagram — Managed-Float Policy Band (S$NEER)



Section J. Singapore's Managed-Float Framework (MAS)

- **Operating target:** S$NEER (trade-weighted basket).
- **Three policy levers: slope** (rate of appreciation), **width** (band tolerance), **centre** (re-centring up or down).
- **Tightening:** steepen slope and/or re-centre upward.
- **Easing:** flatten slope, widen band, or re-centre downward.
- **Why exchange rate, not interest rate?** The impossible trinity — with a free capital account, Singapore must give up independent interest-rate policy. Exchange rate has the largest leverage on $(X - M)$ and on imported inflation given high import content.

Section K. Worked Example

Worked Example: 2022 MAS Tightening

MAS tightened S$NEER five times in 2021–23. Mechanisms: re-centred upward (twice) and steepened slope (twice), increased width (once). S$NEER appreciated ∼8%. SRAS shifted right via lower import costs; AD shifted left via $X \downarrow$, $M \uparrow$. Core inflation peaked at 5.5% in Q1 2023 and moderated to ∼3% by end-2024. Trade-off: exporter margins compressed; tourism more expensive in SGD. Complementary fiscal cushioning via Assurance Package.

NOTE: Frameworks reproduced word-for-word from the Crucible Education Centre exchange-rate handout.